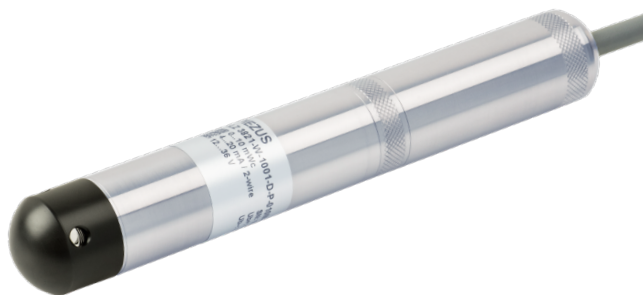


DESCRIPTION

ALZ 3821 is a precision submersible hydrostatic level transmitter with detachable probe, allowing quick and easy removal for service or replacement. High-quality silicon piezoresistive sensor, electronic module with an active temperature compensation technology, robust stainless steel housing and a hydrometric cable with the ventilation tube make ALZ 3821 an outstanding choice for the applications where better accuracy is required.

**SPECIFICATIONS**

Level: 1 mH₂O up to 250 mH₂O

Basic accuracy: $\pm 0.1\%$

Probe: Ø27 mm, AISI 316L, detachable from the cable for easy service

Outputs: 4...20 mA; 0...20 mA; RS-485 / Modbus RTU; HART®

Sensor: silicon piezoresistive

Media temperature: -20...+70 °C

Optional: Ex ia, Pt100 temperature sensor

HART® is a registered trademark of the HART Communication Foundation.

APPLICATIONS

Water, wastewater

Fuels, oils

Tanks, wells

Pump protection

Laboratory equipment

TECHNICAL SPECIFICATIONS

MEASURING RANGES

Pressure range, bar	Level, mH ₂ O	Overpressure, bar	Pressure range, bar	Level, mH ₂ O	Overpressure, bar
0...0.1	0...1	1	0...2.5	0...25	6
0...0.16	0...1.6	1	0...4	0...40	15
0...0.25	0...2.5	1	0...6	0...60	15
0...0.4	0...4	1	0...10	0...100	30
0...0.6	0...6	3	0...16	0...160	60
0...1	0...10	3	0...25	0...250	60
0...1.6	0...16	6	-	-	-

PERFORMANCE	P ≥ 0.1 bar	P < 0.1 bar
Accuracy, % of span*	≤ ±0.1	≤ ±0.2
Temperature effect (% of span / 10 °C)	≤ ±0.02	≤ ±0.04
Compensated range	-20...+70 °C	0...+70 °C
Power supply effect	≤ ±0.05% of span / 10 V	
Load resistance effect	≤ ±0.05% of span / kOhm (transmitters with current output)	
Long-term stability	≤ ±0.1% of span / year	
Response time (10...90%)	≤ 1 ms	

* Accuracy includes non-linearity, hysteresis and non-repeatability.

OPERATING CONDITIONS

Medium temperature (depends on seal)	-20...+70 °C		
Cable service temperature	PVC:	-20...+70 °C	
(static applications)	PUR:	-50...+70 °C	
Storage temperature	-20...+70 °C		
Approval	0Ex ia IIC T6...T4 Ga X		
Temperature class	T4	T5	T6
Ambient temperature	-20...+70 °C	-20...+60 °C	-20...+50 °C
Vibration resistance	10 g RMS, 25–2000 Hz		
Shock resistance	100 g / 11 ms		
Service life	> 100 x 10 ⁶ cycles		

MECHANICAL SPECIFICATIONS

Housing	stainless steel 316L (1.4404)
Seal	FKM; EPDM; NBR
Diaphragm	stainless steel 316L (1.4435)
Cable external jacket	PVC, PUR, FEP
Ingress protection	IP69K

ELECTRICAL SPECIFICATIONS

Output signal	Power supply, U _s	Load resistance, R	Power consumption
4...20 mA / 2-wire	12...36 V	$\leq [(U_s - 12 \text{ V}) / 0.02 \text{ A}] \text{ Ohm}^*$	$\leq 26 \text{ mA}$
4...20 mA / HART®		$\leq 500 \text{ Ohm}$	
4...20 mA / 3-wire	5 V	$\geq 5 \text{ kOhm}$	$\leq 2 \text{ mA}$
0.5...4.5 V / 3-wire	6...15 V		$\leq 7 \text{ mA}$
0.5...4.5 V / 3-wire	12...36 V	-	
RS-485 / Modbus RTU			

* For output signal 4...20 mA / HART® minimum load resistance for digital communication: 250 Ohm.

Safe values for intrinsically safe design 0Ex ia IIC T6...T4 Ga X:

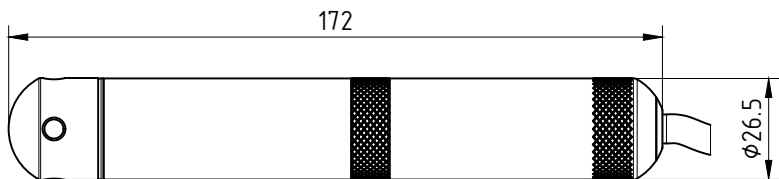
Parameter	2-wire	3-wire
Maximum voltage, U _i	28 V	6 V
Maximum current, I _i	93 mA	60 mA
Maximum power, P _i	660 mW	100 mW
Maximum internal inductance, L _i	10 µH	10 µH
Maximum internal capacitance, C _i	15 nF	500 nF

ELECTRICAL CONNECTIONS / PIN ASSIGNMENT

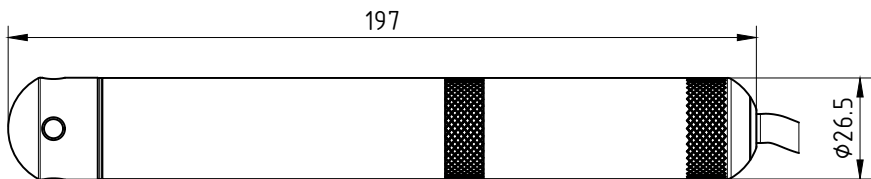
Circuits		Cable gland wires
2-wire	power +	white (red)
	power -	brown (blue)
	ground	yellow-green
3-wire	power +	white (red)
	power -	brown (blue)
	signal +	green (black)
	ground	yellow-green
	T+	yellow
Pt100 (optional)	T-	pink (white)
	T-	gray (green)
RS-485	power +	white (red)
	power -	brown (blue)
	A	yellow
	B	green (black)
	ground	yellow-green
HART®	power +	white
	power -	brown
	ground	yellow-green

DIMENSIONS (mm)

Standard



with HART or Ex ia



ORDERING CODE

ALZ 3821		-X	-XXXX	-X	-X	-X	-XX	// X	-XXXX
UNIT OF MEASUREMENT									
	bar	B							
	kg/cm ²	S							
	mH ₂ O	W							
	kPa	K							
	Other (specify when ordering)	X							
UPPER RANGE LIMIT (URL)									
bar, kg/cm ²	mH ₂ O	kPa							
0.1	0100	1	1000	10	1001				
0.16	0160	1.6	1600	16	1601				
0.25	0250	2.5	2500	25	2501				
0.4	0400	4	4000	40	4001				
0.6	0600	6	6000	60	6001				
1	1000	10	1001	100	1002				
1.6	1600	16	1601	160	1602				
2.5	2500	25	2501	250	2502				
4	4000	40	4001	400	4002				
6	6000	60	6001	600	6002				
10	1001	100	1002	1000	1003				
16	1601	160	1602	Other	XXXX				
25	2501	250	2502						
Other	XXXX	Other	XXXX						
ACCURACY									
	0.1% (P > 0.1 bar) (standard)	A							
	0.2% (P ≤ 0.1 bar) (standard)	B							
	Other (specify when ordering)	X							
OUTPUT SIGNAL									
	4...20 mA / 2-wire (standard)	A							
	4...20 mA / 2-wire, 0Ex ia IIC T6...T4 Ga X	Q							
	4...20 mA / 3-wire	B							
	0.5...4.5 V / 3-wire, U _s = 5 V, 0Ex ia IIC T6...T4 Ga X	R							
	0.5...4.5 V / 3-wire, U _s = 6...15 V	K							
	RS-485 / Modbus RTU	M							
	4...20 mA / HART®	H							
	Other (specify when ordering)	X							
SEALS									
	FKM	F							
	NBR	N							
	EPDM	E							
	Other (specify when ordering)	X							
VERSION									
	Standard	00							
	Pt100 temperature sensor	0T							
	Other (specify when ordering)	XX							

ORDERING CODE (CONTINUED)

ALZ 3821	-X	-XXXX	-X	-X	-X	-XX	// X	-XXXX
CABLE EXTERNAL JACKET								
							PVC	P
							PUR	U
							FEP	T
							Other (specify when ordering)	X
CABLE LENGTH								
Any length in meters (e.g., 010M means 10 meters)								XXXM

Example: ALZ 3821-W-6000-A-A-F-00 // P-010M

ACCESSORIES

				
Anchoring clamp for submersible transmitter	PZ 1024 Power supply unit	BZ 05 / BZ 10 Dry air junction box for submersible transmitters		